

Structures and File Handling

1. P01

Define a struct Date (day, month, year) and the following functions.

1. Input a date (from text file).
2. Output a date (yyyy-MM-dd) (to text file).
3. Output a date with a given format (to text file). For example: yy: 70, yyyy: 1970, M: 8 or 12, MM: 08 or 12, MMM: Jan ... Dec, d: 1 or 29, dd: 01 or 29.

```
outputDateWithFormat(d1, "yyyy-MM-dd"); // 2019 - 12 - 04
outputDateWithFormat(d1, "MMM dd yy"); // Dec 12 19
```

4. Compare 2 date.
5. Find tomorrow: increase a date by 1 day.
6. Find yesterday: decrease a date by 1 day.
7. Increase a date by k day.
8. Decrease a date by k day.
9. Compute the distance between input date vs 1/1/same year.
10. Compute the distance between input date vs 1/1/1970.
11. Compute the distance between 2 input dates.
12. Given that the date of week of 1970-01-01 is Thursday. Compute the date of week of the input date.

2. P02

Define a struct Student (id, full name, gpa, address, date of birth) and the following functions.

15. Input a student (from text file).
16. Output a student: id, full name, gpa, address, dob (to text file).
17. Output a student: id, first name (first word), last name (remaining words), gpa, address, dob, age (18 year 2 months, 2 days) (to text file).
18. Extract the class of input student (to text file). For example, if your id is 19127001, you are in K19 class.
19. Compare 2 students by id.
20. Compare 2 students by gpa, and then by id.
21. Compare 2 students by name, and then by id.
22. Compare 2 students by first name, and then by id.
23. Compare 2 students by last name, and then by id.
24. Compare 2 students by dob, and then by id.

3. P03

Define a struct Course (course id, e.g. CSC10001, course name, e.g. Introduction to Programming, and an array of students, status, e.g. open, or close, maximum number of students in the course, minimum number of students in the course) and the following functions.

27. Input a course (from text file).
28. Output a course: id, name, students' info (to text file)
29. Add a student to a course. Check the maximum number of students in the course condition.
30. Remove a student to a course. Check minimum number of students in the course to update the course status: open or close.
31. Get the current date in your computer. Find all students who were born in this month and print the result.
32. Get the current date in your computer. Find all students who were born in this date and print the result.
33. Get the current date in your computer. Find all students who are legal to have driving licences (≥ 18 years old) print the result.
34. Find all students are in K19 class and print the result (to text file).
35. Find a student by a given id and print the result (to text file).
36. Find a student by a given name ("thanh" vs "Tuan-Thanh Ho"). Then print the result (to text file).
37. Sort students by id. Then print the result (to text file).
38. Sort students by first name. Then print the result (to text file).
39. Sort students by gpa. Then print the result (to text file).
40. Sort students by dob. Then print the result (to text file).